

4(a)	any three from: 1 ref. to isolate / extract, DNA, from, cells / tissue / named ; 2 ref. to cut DNA, into small fragments / using restriction enzymes ; 3 ref. to denature / AW, into single-stranded DNA ; 4 add fluorescent, tag / marker / dye ;	3
4(b)(i)	2 400 000 or 2 500 000 ;	1
4(b)(ii)	any three from: 1 DNA from DiGeorge syndrome and control DNA labelled with different colour fluorescent tags ; 2 <u>DNA</u> hybridises with probes (on microarray) ; <i>DiGeorge Syndrome</i> 3 less DNA binds to probes (than control DNA) ; 4 so there is, less / 50%, fluorescence where nucleotides have been deleted ; 5 fluorescence is, equal / 100%, where nucleotides have not been deleted ;	3

Question	Answer	Marks
4(b)(iii)	any three from: 1 different number of nucleotides may be deleted ; 2 different genes may be deleted ; 3 an individual only has one allele (of the deleted genes) ; 4 different individuals may have different alleles of the same gene or in one individual remaining allele may be recessive but dominant in another individual ; 5 so different proteins made ; 6 ref. to individuals may have different environments ; A diet	3